Physiologically-Constrained Neural Operators
for Heart Sound Classification

Comprehensive Lecture Notes
Fourier Neural Operator vs. Mel-Spectrogram CNN Baseline

"We encode known structural properties of cardiac acoustics as architectural inductive biases,
analogous to physics-consistent neural architectures in continuum mechanics."

Topics Covered

- Cardiac acoustics and phonocardiography (PCG)
- Fourier Neural Operators (FNO) from first principles
- Physiological frequency masking (S1/S2 sounds)
- Physics-informed loss functions
- Mel-spectrogram CNN baseline architecture
- t-SNE latent space analysis
- Comparative evaluation and interpretability

Prepared from notebook: `heart_sound_fno_classifier.ipynb`

Contents

1. Introduction and Clinical Motivation

Valvular heart disease is among the leading causes of cardiovascular morbidity worldwide. Auscultation—the clinical practice of listening to heart sounds through a stethoscope—has been used for over two centuries to detect valvular abnormalities. However, accurate diagnosis from auscultation requires years of clinical training, and inter-rater agreement is often low. Automated classification of phonocardiogram (PCG) signals promises to democratize screening, enabling reliable detection in resource-limited settings and at scale.

This notebook develops and compares two deep learning architectures for five-class heart sound classification. Rather than treating the problem as a purely statistical pattern recognition task, the primary model—a **Fourier Neural Operator (FNO)**—encodes known physiological structure of cardiac acoustics directly into its architecture and training objective. This philosophy is directly analogous to the use of physics-informed neural networks (PINNs), variational constraints, and polyconvexity conditions in computational mechanics.

1.1 The Five Classes

The dataset contains recordings from five clinical categories:

Label	Full Name	Clinical Description
AS	Aortic Stenosis	Narrowing of the aortic valve; systolic ejection murmur
MR	Mitral Regurgitation	Leaky mitral valve; holosystolic murmur
MS	Mitral Stenosis	Narrowing of the mitral valve; low-pitched diastolic murmur
MVP	Mitral Valve Prolapse	Billowing of mitral leaflets; mid-systolic click
N	Normal	Healthy heart sounds; S1 and S2 only

1.2 The Phonocardiogram Signal

A PCG signal $s(t)$ is a one-dimensional time series recording acoustic pressure variations at the chest wall. Each cardiac cycle contains two dominant sounds:

- **S1 (the "lub")** — caused by closure of the mitral and tricuspid valves at the beginning of systole. Energy is concentrated in the 25–45 Hz band.
- **S2 (the "dub")** — caused by closure of the aortic and pulmonic valves at the beginning of diastole. Energy is concentrated in the 50–70 Hz band.

Pathological conditions alter the timing, intensity, duration, and frequency content of these sounds and introduce additional murmurs between them. This frequency-domain structure is the key physiological prior that the notebook exploits.

2. Dataset and Preprocessing

2.1 Data Pipeline

Each recording is a `.wav` file stored in one of five class-labeled subdirectories. The dataset loading pipeline applies three operations uniformly:

1. **Resampling.** All recordings are resampled to a common sample rate $f_s = 22{,}050\,\text{Hz}$ using `librosa`'s high-quality resampler. This standardizes the frequency axis across recordings that may have been acquired at different rates.

2. **Fixed-length windowing.** Each signal is either *trimmed* (if longer than $L = 66{,}150$ samples $= 3.0\,\text{s}$) or *zero-padded on the right* (if shorter). This produces a uniform tensor shape $(1, L)$ required for batched processing.
   Remark 2.1. The dataset has a wide duration distribution: minimum 1.16 s, maximum 3.99 s, mean $2.44 \pm 0.36\,\text{s}$. A window of 3.0 s means the majority of recordings are zero-padded. A more conservative choice would be 2.5 s; the 3 s window was chosen to avoid truncating longer, clinically informative recordings.

3. **Amplitude normalization.** Each waveform is divided by its peak absolute amplitude, mapping it into $[-1, 1]$. This removes gain variability between recording devices and microphone placements:
$$\tilde{s}(t) = \frac{s(t)}{\max_t |s(t)|}$$

2.2 Train / Validation / Test Split

The full dataset is partitioned using a fixed random seed (42) into three non-overlapping subsets:

$$N_{\text{train}} = N_{\text{total}} - N_{\text{val}} - N_{\text{test}}, \quad N_{\text{val}} = \lfloor 0.15\, N_{\text{total}} \rfloor, \quad N_{\text{test}} = \lfloor 0.15\, N_{\text{total}} \rfloor.$$

PyTorch's `random_split` handles this with a `Generator` seeded at 42 for reproducibility. The test set is held out entirely during training and hyperparameter selection; it is only accessed for final evaluation.

3. Architecture I: The Fourier Neural Operator (FNO)

3.1 Conceptual Motivation: Why Fourier Space?

Standard convolutional neural networks operate in the physical (time) domain. They apply local kernels of fixed receptive field, which makes them efficient for detecting local features but potentially inefficient at capturing global, periodic structure.

Cardiac PCG signals are inherently *periodic*. A healthy heart at 60–80 BPM produces a signal whose dominant energy concentrations repeat at regular intervals. Moreover, the physiologically meaningful features—S1 and S2 sounds—are *frequency-band-specific*. Working directly in the Fourier domain is therefore a natural inductive bias.

The Fourier Neural Operator (FNO), introduced by Li et al. (2021), parameterizes *integral operators in Fourier space*. The core insight is that any convolution operator in the spatial domain corresponds to pointwise multiplication in the Fourier domain:

$$(\mathcal{K} \cdot u)(x) = \int k(x-y)\, u(y)\, dy \quad \Longleftrightarrow \quad \widehat{(\mathcal{K} \cdot u)}(\xi) = \widehat{k}(\xi)\, \widehat{u}(\xi).$$

The FNO learns the operator $\widehat{k}(\xi)$ directly in Fourier space, truncated to the M most informative modes.

3.2 Overall FNO Architecture

The full `FNOClassifier` model consists of four stages:

Stage	Module	Purpose
1	`PhysioFrequencyMask`	Physiological band-pass prior
2	`Conv1d(1, W, 1)` (lift)	Project scalar signal to W-dim feature space
3	D × `FNOBlock1d`	Spectral feature learning
4	Global avg. pool + MLP head	Aggregate and classify

With hyperparameters $W = 64$ (width), $M = 20$ (modes), $D = 4$ (depth).

3.3 The Spectral Convolution Layer

This is the fundamental building block of the FNO. It learns complex-valued weights $\widehat{W} \in \mathbb{C}^{C_{\text{in}} \times C_{\text{out}} \times M}$ on the M retained Fourier modes, then maps the input back to the time domain.

Given an input tensor $\mathbf{x} \in \mathbb{R}^{B \times C_{\text{in}} \times L}$ (batch × channels × length):

1. Compute the real FFT:

$$\hat{\mathbf{x}} = \texttt{rfft}(\mathbf{x}) \in \mathbb{C}^{B \times C_{\text{in}} \times (L/2+1)}$$

2. Retain only the M lowest-frequency modes and apply the learned operator:

$$\hat{\mathbf{y}}_{b,o,k} = \sum_{i=1}^{C_{\text{in}}} \hat{\mathbf{x}}_{b,i,k}\, \hat{W}_{i,o,k}, \quad k = 0, 1, \ldots, M-1$$

   where $\hat{W} = W_r + i\, W_i$ with real-valued learnable parameters W_r, W_i (parameterizing as two real tensors avoids complex-number gradient issues).

3. Zero-fill the high-frequency modes and invert:

$$\mathbf{y} = \texttt{irfft}(\hat{\mathbf{y}}, n = L) \in \mathbb{R}^{B \times C_{\text{out}} \times L}$$

The use of `norm='ortho'` in both FFT directions ensures the transform is unitary, preserving energy scaling across layers.

3.4 The FNO Block

Each `FNOBlock1d` combines the spectral convolution with a bypass (residual) path, analogous to a ResNet block but in Fourier space:

$$\mathbf{x}^{(\ell+1)} = \mathrm{GELU}\Big(\texttt{InstanceNorm}\big(\underbrace{\mathcal{K}_\ell \cdot \mathbf{x}^{(\ell)}}_{\text{spectral path}} + \underbrace{W_\ell \, \mathbf{x}^{(\ell)}}_{\text{local bypass}}\big)\Big)$$

where $\mathcal{K}_\ell$ is the spectral convolution and W_ℓ is a pointwise $1{\times}1$ convolution. This design is critical:

- The **spectral path** captures long-range, global, periodic interactions across the full signal duration.
- The **bypass path** captures local, instantaneous channel interactions, providing a complementary inductive bias.
- **InstanceNorm** normalizes each channel independently per sample, suitable for variable-length signals (more appropriate than BatchNorm for time series).
- **GELU** (Gaussian Error Linear Unit) $= x\,\Phi(x)$ is smooth and has been shown to outperform ReLU in deep sequence models.

3.5 The Classifier Head

After D FNO blocks, the output has shape (B, W, L). **Global average pooling** over the time dimension collapses this to (B, W):

$$\bar{\mathbf{x}} = \frac{1}{L}\sum_{\ell=1}^{L} \mathbf{x}_\ell \in \mathbb{R}^{B\times W}$$

This operation is *length-invariant*, meaning the classifier is robust to the variable signal lengths present after zero-padding. The pooled representation is then passed through a two-layer MLP:

$$\mathbf{z} = \mathrm{GELU}(W_2 \; \mathrm{GELU}(W_1\,\bar{\mathbf{x}} + b_1) + b_2) \in \mathbb{R}^{B\times 64}$$

with dropout ($p = 0.3$) for regularization, followed by a linear classifier:

$$\hat{\mathbf{y}} = W_c\,\mathbf{z} + b_c \in \mathbb{R}^{B\times 5}$$

The vector $\mathbf{z} \in \mathbb{R}^{64}$ is the **latent embedding** returned alongside the logits, and is used later for t-SNE visualization.

4. Physiological Frequency Masking (S1/S2)

4.1 What Is It?

The `PhysioFrequencyMask` module is a *learnable soft band-pass filter* that is applied to the raw input signal before any FNO processing. It is the first architectural element encountered by every input sample.

The known frequency anatomy of heart sounds provides a strong prior:

- **S1** (first heart sound): energy in 25–45 Hz
- **S2** (second heart sound): energy in 50–70 Hz
- **Murmurs**: may extend up to 400–600 Hz
- **Ambient noise / artifacts**: dominant at very high and very low frequencies

Initializing the mask to up-weight the S1/S2 bands before any learning begins is analogous to initializing a neural network from a physically meaningful solution rather than from random weights.

4.2 Mathematical Formulation

Let $f_k = k \cdot f_s / L$ denote the frequency of the k-th Fourier mode ($k = 0, 1, \ldots, M-1$). Define the initial mask:

$$m_k^{(0)} = \begin{cases} 1.0 & \text{if } f_k \in [25, 45]\,\text{Hz (S1 band) or } f_k \in [50, 70]\,\text{Hz (S2 band)} \\ 0.1 & \text{otherwise} \end{cases}$$

To ensure the mask remains in $(0, 1)$ during training, the parameters are stored in *logit space*:

$$\theta_k = \ln\left(\frac{m_k^{(0)}}{1 - m_k^{(0)}}\right) \quad \Longrightarrow \quad m_k = \text{sigmoid}(\theta_k) = \frac{1}{1 + e^{-\theta_k}}$$

The mask is applied in the Fourier domain:

$$\tilde{s}(\xi) = m(\xi) \cdot \hat{s}(\xi)$$

$$x_{\text{masked}}(t) = \texttt{irfft}(\tilde{s}, n = L)$$

where modes beyond the M retained ones retain a small baseline weight of 0.1 (not zero, to preserve gradient flow).

4.3 What the Mask Learns

Initially, the mask is a hard physiological prior. During training, gradients from the classification and physiological loss functions flow back through the mask parameters θ_k, allowing the mask to *adapt*:

- If a mode outside the S1/S2 bands turns out to be discriminative (e.g., a murmur frequency specific to one class), the mask weight for that mode will increase from 0.1 toward 1.0.
- Modes carrying only noise will remain suppressed.
- The final mask, visualized in Section 10 of the notebook, reveals which frequencies the model has found important—providing *interpretability*.

This is the key advantage over a fixed filter bank: the physiological prior guides initialization, but the mask is not locked—it can refine itself using data.

5. Architecture II: CNN Baseline (Mel Spectrogram)

5.1 Purpose and Role

The `CNNBaseline` is not a competitor in the sense of being the model we want to deploy. Its role is to serve as an *ablation reference*: a strong, conventional architecture that does *not* incorporate any physiological priors. Comparing the two models answers the question: *how much does the physiological inductive bias actually help?*

5.2 The Mel Spectrogram Front-End

Rather than processing the raw waveform, the CNN baseline first converts the signal to a *log-Mel spectrogram*—a 2D time-frequency representation that matches human auditory perception. This is implemented entirely in PyTorch (no librosa at inference time) via a `MelSpectrogramLayer`.

5.2.1 Short-Time Fourier Transform (STFT)

The STFT divides the signal into overlapping windows and computes the DFT of each:

$$X(m,k) = \sum_{n=0}^{N_{\text{FFT}}-1} s(n+m\cdot h)\, w(n)\, e^{-2\pi i k n/N_{\text{FFT}}}$$

where $w(n)$ is a Hann window of size $N_{\text{FFT}} = 512$, $h = 128$ is the hop length, and m, k index time frame and frequency bin respectively. The power spectrogram is $|X(m,k)|^2$.

5.2.2 Mel Filterbank

The Mel scale is a perceptual frequency scale that mimics the non-linear sensitivity of the human cochlea. The conversion between Hz and Mel is:

$$f_{\text{mel}} = 2595 \log_{10}\left(1 + \frac{f_{\text{Hz}}}{700}\right)$$

$N_{\text{mels}} = 64$ triangular filters are uniformly spaced on the Mel scale between $f_{\min} = 20\,\text{Hz}$ and $f_{\max} = f_s/2 = 11{,}025\,\text{Hz}$. The filterbank matrix $\mathbf{F} \in \mathbb{R}^{N_{\text{mels}}\times(N_{\text{FFT}}/2+1)}$ maps the power spectrum to Mel-frequency bins.

5.2.3 Log Compression

$$S_{\text{mel}}(m,k) = \log\Big(\mathbf{F}\cdot|X(m,\cdot)|^2 + \varepsilon\Big), \quad \varepsilon = 10^{-6}$$

The log compression compresses the dynamic range, making quiet sounds and loud murmurs comparably represented.

5.3 CNN Backbone

The Mel spectrogram $S_{\text{mel}} \in \mathbb{R}^{B\times 1\times N_{\text{mels}}\times T}$ is processed by three 2D convolutional blocks:

Layer	Output channels	Operation
Conv Block 1	32	Conv2d(3×3) + BatchNorm + ReLU + MaxPool(2×2)
Conv Block 2	64	Conv2d(3×3) + BatchNorm + ReLU + MaxPool(2×2)
Conv Block 3	128	Conv2d(3×3) + BatchNorm + ReLU + AdaptiveAvgPool(4×4)

The adaptive average pooling to $(4, 4)$ makes the architecture resolution-invariant. A two-layer MLP head ($128\times16 \to 256 \to 5$) with Dropout(0.4) produces the logits.

5.4 Why Two Architectures?

Aspect	FNO Classifier	CNN Baseline
Input representation	Raw waveform (1-D)	Log-Mel spectrogram (2-D)
Domain	Fourier (spectral)	Time-frequency (spatial)
Physiological priors	Yes — mask, periodicity, hierarchy	No
Long-range dependencies	Global (full signal FFT)	Local (3×3 kernel)
Interpretability	Learned frequency mask	Limited
Training loss	Three-term (CE + physio)	Cross-entropy only

The CNN represents the de facto standard approach in audio classification; the FNO represents the physics-informed approach. Their comparison is the central scientific question of the notebook.

6. Physiological Loss Functions

6.1 Overview of the Loss Design

The FNO is trained with a composite loss function:

$$\boxed{\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda_1\,\mathcal{L}_{\text{periodicity}} + \lambda_2\,\mathcal{L}_{\text{freq hierarchy}}}$$

with $\lambda_1 = \lambda_2 = 0.1$ in the notebook.

Each term encodes a distinct physiological belief about what PCG signals should look like.

6.2 Cross-Entropy Loss $\mathcal{L}_{\text{CE}}$

6.2.1 Definition

For a batch of B samples with logits $\hat{\mathbf{y}} \in \mathbb{R}^{B\times 5}$ and integer labels $y_b \in \{0,1,2,3,4\}$:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{B}\sum_{b=1}^{B} \log\left(\frac{\exp(\hat{y}_{b,y_b})}{\sum_{c=0}^{4}\exp(\hat{y}_{b,c})}\right)$$

6.2.2 Why Cross-Entropy?

Cross-entropy is the natural loss for classification because it is equivalent to maximum likelihood estimation under a categorical distribution. Minimizing $\mathcal{L}_{\text{CE}}$ drives the softmax output probabilities toward the one-hot label vector. It penalizes confident wrong predictions much more harshly than uncertain ones (a desirable property for clinical applications where false negatives can be dangerous).

6.3 Periodicity Loss $\mathcal{L}_{\text{periodicity}}$

6.3.1 Physiological Motivation

The healthy heart (and many pathological hearts) beats at a regular rate. The PCG signal therefore has energy concentrated near the fundamental frequency of the heart rate (HR) and its harmonics:

$$f_{\text{HR}} \approx \frac{60\text{ BPM}}{60\text{ s/min}} \approx 1\text{ Hz} \quad \text{to} \quad f_{\text{HR}} \approx \frac{80\text{ BPM}}{60} \approx 1.33\text{ Hz}$$

with harmonics at $2f_{\text{HR}}, 3f_{\text{HR}}, 4f_{\text{HR}}$, covering up to approximately $5.3\,\text{Hz}$.

6.3.2 Mathematical Formulation

Let $P_b(\xi) = |\hat{s}_b(\xi)|^2$ denote the power spectral density of signal b. Define the cardiac frequency band:

$$\mathcal{B}_{\text{HR}} = \{\xi : 1.0\,\text{Hz} \le \xi \le 5.3\,\text{Hz}\}$$

The periodicity loss is the fraction of total energy *outside* this band, averaged over the batch:

$$\mathcal{L}_{\text{periodicity}} = 1 - \frac{1}{B}\sum_{b=1}^{B}\frac{\sum_{\xi\in\mathcal{B}_{\text{HR}}} P_b(\xi)}{\sum_{\xi} P_b(\xi) + \varepsilon}$$

6.3.3 Interpretation

- $\mathcal{L}_{\text{periodicity}} = 0$ means all spectral energy lies in the cardiac HR band — perfectly periodic.
- $\mathcal{L}_{\text{periodicity}} = 1$ means no energy in the HR band — completely aperiodic.
- By minimizing this loss alongside $\mathcal{L}_{\text{CE}}$, the network is regularized to attend to the periodic structure of the cardiac cycle, rather than fitting noise or aperiodic artifacts.

6.3.4 Why Not Use This Loss Alone?

Periodicity loss is not discriminative by itself — all five classes have quasi-periodic structure. It acts as a *regularizer*: it constrains the solution space without driving the decision boundary. The classification information comes from $\mathcal{L}_{\text{CE}}$.

6.4 Frequency Hierarchy Loss $\mathcal{L}_{\text{freq hierarchy}}$

6.4.1 Physiological Motivation

Across all heart sound classes (including pathological ones), the S1 sound is acoustically louder and carries more energy than S2. This is a physiological fact: the mitral/tricuspid valve closure (S1) generates a more forceful acoustic event than the semilunar valve closure (S2). Therefore, for any valid PCG recording:

$$\text{Energy in S1 band} \geq \text{Energy in S2 band}$$

6.4.2 Mathematical Formulation

Let:

$$E_{\text{S1}}^{(b)} = \text{mean}_{k \in [25,45]\,\text{Hz}} P_b(f_k), \qquad E_{\text{S2}}^{(b)} = \text{mean}_{k \in [50,70]\,\text{Hz}} P_b(f_k)$$

The frequency hierarchy loss is a *hinge loss* that penalizes violations of the S1 $\geq$ S2 energy constraint:

$$\boxed{\mathcal{L}_{\text{freq hierarchy}} = \frac{1}{B} \sum_{b=1}^{B} \text{ReLU}\big(E_{\text{S2}}^{(b)} - E_{\text{S1}}^{(b)}\big)}$$

6.4.3 Why a Hinge (ReLU) Form?

The ReLU form $\text{ReLU}(E_{\text{S2}} - E_{\text{S1}})$ is zero when the constraint $E_{\text{S1}} \geq E_{\text{S2}}$ is satisfied, and increases linearly in the degree of violation. This is exactly the hinge loss structure used in constrained optimization:

- It applies **no gradient** when the constraint is already satisfied (no penalty for correct physiology).
- It applies **a constant gradient** proportional to the violation when the constraint is broken.
- It is **differentiable everywhere except at zero**, making it compatible with gradient-based optimization.

Remark 6.1. This is directly analogous to how inequality constraints are handled in mechanics. In continuum mechanics, conditions such as the Legendre-Hadamard inequality for strong ellipticity or polyconvexity of the strain energy density can be enforced as soft constraints in a loss function using penalty or hinge formulations. The frequency hierarchy loss is the acoustic analog: a known physical inequality encoded as a penalty.

6.5 The λ Weighting and Balance

The weights $\lambda_1 = \lambda_2 = 0.1$ serve to balance the scales of the three losses. If $\mathcal{L}_{\text{CE}} \sim 1.0$ (typical early training), and the physiological losses are also $O(1)$, then $\lambda = 0.1$ means the physiological terms contribute $\sim 10\%$ of the gradient magnitude. This is intentional: the constraints *guide* the optimization, but do not dominate it and override the discriminative signal.

7. Training Procedure

7.1 Optimizer: AdamW

Both models are trained with AdamW:

$$\theta_{t+1} = \theta_t - \alpha\, \hat{m}_t / (\sqrt{\hat{v}_t} + \varepsilon) - \alpha \lambda_{\text{wd}}\, \theta_t$$

where $\hat{m}_t, \hat{v}_t$ are bias-corrected first and second moment estimates of the gradient, $\alpha = 10^{-3}$ is the learning rate, and $\lambda_{\text{wd}} = 10^{-4}$ is the weight decay coefficient. The decoupled weight decay in AdamW (as opposed to the L2 penalty in Adam) provides better regularization properties.

7.2 Learning Rate Schedule: Cosine Annealing

The learning rate follows a cosine decay over $T = 60$ epochs:

$$\alpha_t = \alpha_{\min} + \frac{1}{2}(\alpha_{\max} - \alpha_{\min}) \left(1 + \cos\left(\frac{\pi t}{T}\right)\right)$$

This smoothly reduces the learning rate, allowing the optimizer to make large updates early in training and fine-grained updates near convergence.

7.3 Gradient Clipping

Gradients are clipped by global ℓ^2 norm to 1.0 before each optimizer step:

$$g \leftarrow g \cdot \frac{1.0}{\max(1.0, \|g\|_2)}$$

This prevents gradient explosions that can occur in spectral operations when the Fourier coefficients have large magnitudes.

7.4 Model Checkpointing

The model weights corresponding to the best validation accuracy are saved at each epoch. After training completes, the best checkpoint is restored for final evaluation. This is standard practice to avoid selecting overfitted weights from the final epoch.

8. Evaluation: Learned Physiological Frequency Mask

8.1 What Is Being Visualized?

After training, the `PhysioFrequencyMask` module has adapted its logit parameters $\{\theta_k\}_{k=0}^{M-1}$. The mask weights $m_k = \text{sigmoid}(\theta_k)$ can be read out and plotted against the corresponding physical frequencies f_k (in Hz).

8.2 How to Interpret the Plot

The learned mask plot (Section 10 of the notebook) shows:

- **Bar heights** (in $[0, 1]$): the learned weight for each retained Fourier mode. A weight near 1.0 means the network has decided this frequency is important for classification. A weight near 0.1 (the initialization for non-S1/S2 modes) means it has been largely ignored.
- **Blue shaded region**: the initialized S1 band (25–45 Hz).
- **Red shaded region**: the initialized S2 band (50–70 Hz).

Three possible outcomes reveal different stories about the data:

1. **S1/S2 weights remain high, others stay low**: the physiological prior was correct and the data confirms it. The model is doing what we intended.
2. **Some non-S1/S2 weights grew**: those frequencies (e.g., murmur bands) contain discriminative information. The model has discovered this automatically.
3. **S1/S2 weights dropped**: the initialization prior was overridden by the data. This would be surprising and would warrant investigation of the recording quality.

8.3 The Mask as an Interpretability Tool

The learned mask provides a form of *post-hoc interpretability without requiring separate attribution methods* (like SHAP or Grad-CAM). Because the mask directly weights Fourier modes in physical frequency units, clinicians can inspect which frequency bands drove the classification decision. This is a direct benefit of the physics-informed architectural design.

9. Evaluation: t-SNE of FNO Latent Embeddings

9.1 What Are Latent Embeddings?

Recall from Section 3.5 that the FNO classifier returns two outputs: the logits $\hat{\mathbf{y}}$ and the embedding $\mathbf{z} \in \mathbb{R}^{64}$. The embedding is the 64-dimensional vector produced by the projection MLP before the final linear classifier. It is the model's internal *compressed representation* of the heart sound — a point in a 64-dimensional space where classification is performed by a linear boundary.

9.2 t-SNE Dimensionality Reduction

64 dimensions cannot be directly visualized. t-SNE (t-distributed Stochastic Neighbor Embedding, van der Maaten & Hinton 2008) reduces the embeddings to 2D while approximately preserving *local neighborhood structure*:

1. For each point i, define a probability distribution over neighbors in 64-D space using a Gaussian kernel:
$$p_{j|i} = \frac{\exp(-\|z_i - z_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|z_i - z_k\|^2 / 2\sigma_i^2)}$$

2. In 2-D, define a similar distribution using a Student t-distribution (heavier tails):
$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l}(1 + \|y_k - y_l\|^2)^{-1}}$$

3. Minimize the Kullback-Leibler divergence $\mathrm{KL}(P\|Q)$ via gradient descent in 2-D space. The parameter `perplexity = 15` controls the effective neighborhood size.

9.3 How to Interpret the t-SNE Plot

The t-SNE visualization answers the question: *has the FNO learned to group similar heart sounds together in its internal representation?*

- **Well-separated clusters** (one per class, minimal overlap) indicate that the model has learned class-discriminative representations. Linear classification in the embedding space should be easy, which is consistent with high test accuracy.
- **Overlapping clusters** between confusable classes (e.g., MS and MVP, which can both present with diastolic murmurs) indicate that the model finds these classes intrinsically similar — a clinically reasonable finding.
- **Normal clearly separated from all pathologies**: the most clinically important separation. If the model correctly identifies normal hearts regardless of which pathology is present, it is useful as a screening tool.

Remark 9.1. t-SNE is a visualization tool, not an evaluation metric. Cluster separation in t-SNE does not guarantee linear separability in the full 64-D space, and the axes have no physical meaning. Always use quantitative metrics (accuracy, F1, confusion matrix) as the primary evaluation.

10. Confusion Matrices and Classification Metrics

10.1 Confusion Matrix

The normalized confusion matrix $C \in [0,1]^{5\times 5}$ has entries:
$$C_{ij} = \frac{\text{Number of samples with true class } i \text{ predicted as class } j}{\text{Number of samples with true class } i}$$

Diagonal entries C_{ii} are per-class recall (sensitivity). Off-diagonal entries reveal systematic confusions. Clinically, false negatives (pathology predicted as Normal) are more dangerous than false positives.

10.2 Per-Class Metrics

The `classification_report` provides:
$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c}, \quad \text{F1}_c = \frac{2 \cdot \text{Prec}_c \cdot \text{Rec}_c}{\text{Prec}_c + \text{Rec}_c}$$

The macro-averaged F1 (unweighted average across all classes) is the most informative single number for imbalanced datasets, as it treats all classes equally regardless of their size.

11. Summary of Design Choices and Their Rationale

Design Choice	Implementation	Rationale
Fourier spectral conv.	`SpectralConv1d`	Heart sounds are periodic; FFT is the natural basis
Mode truncation to $M = 20$	Retain low-freq modes only	Cardiac acoustics < 1 kHz; high modes are noise
Physio frequency mask	`PhysioFrequencyMask`	Encode S1/S2 band prior; let data refine it
Logit-space mask param.	$\theta = \log(m/(1-m))$	Guarantees mask $\in (0, 1)$ without projection
Bypass 1×1 conv.	`nn.Conv1d(W,W,1)`	Local features complement global spectral features
InstanceNorm	Per-sample, per-channel	Robust to variable-amplitude and variable-length signals
Global average pooling	`x.mean(dim=-1)`	Length-invariant aggregation after zero-padding
Periodicity loss	Fraction of energy outside HR band	Encodes cardiac rhythmicity as a constraint
Hierarchy loss	$\mathrm{ReLU}(E_{S2} - E_{S1})$	Encodes S1 $\geq$ S2 energy as a hinge constraint
AdamW + cosine LR	Standard + weight decay	Better generalization than Adam alone
Gradient clipping	$\|g\|_2 \leq 1.0$	Prevents spectral explosion in FFT layers

12. Connection to Physics-Informed Machine Learning

The design philosophy of this notebook is directly analogous to approaches in scientific machine learning (SciML) and physics-informed neural networks (PINNs):

1. **Architectural inductive biases**: Just as PINNs for PDEs use automatic differentiation to enforce governing equations in the architecture, the FNO uses spectral convolutions to enforce the periodicity and frequency-band structure of cardiac acoustics.
2. **Soft constraint losses**: Just as polyconvexity or Legendre-Hadamard conditions can be enforced via penalty terms in hyperelastic network training, the periodicity and frequency hierarchy losses enforce physiological inequalities during optimization.
3. **Physics-guided initialization**: Just as initializing a neural network for constitutive modeling near a known analytical solution (e.g., neo-Hookean) accelerates convergence, the S1/S2 mask initialization provides a physiologically meaningful starting point.
4. **Interpretable latent space**: The learned frequency mask provides direct physical interpretability, analogous to how learned material parameters in data-driven constitutive

models can be mapped back to measurable physical quantities.

The notebook's epigraph captures this philosophy succinctly: structural properties of the physical domain (cardiac acoustics) are encoded as *architectural inductive biases*, not merely hoped for as emergent phenomena from data alone.

13. Glossary of Key Terms

PCG (Phonocardiogram)
A digital recording of acoustic heart sounds, typically sampled at $\geq 4000\,\text{Hz}$.

S1 First heart sound; mitral/tricuspid valve closure; 25–45 Hz band.

S2 Second heart sound; aortic/pulmonic valve closure; 50–70 Hz band.

FNO (Fourier Neural Operator)
A neural network architecture that learns operators in Fourier space by applying complex-valued weights to retained frequency modes.

Spectral convolution
Convolution implemented as pointwise multiplication in the Fourier domain, followed by inverse FFT.

Mode truncation
Retaining only the M lowest-frequency Fourier modes, enforcing a band-limited inductive bias.

PhysioFrequencyMask
A learnable soft band-pass filter initialized from S1/S2 physiology and parameterized in logit space.

Inductive bias
A structural assumption built into a model's architecture that constrains the function class it can represent, often improving generalization.

Cross-entropy loss
The standard classification loss, equal to negative log-likelihood under a categorical distribution.

Periodicity loss
A loss term penalizing signals whose energy does not concentrate near the cardiac heart rate fundamental and its harmonics.

Frequency hierarchy loss
A hinge loss enforcing the physiological constraint that S1 band energy exceeds S2 band energy.

Mel spectrogram
A log-power spectrogram on the Mel frequency scale, which mimics human auditory perception.

GELU
Gaussian Error Linear Unit activation: $\mathrm{GELU}(x) = x\,\Phi(x)$ where Φ is the Gaussian CDF.

InstanceNorm
Normalization of each channel independently per sample, robust to variable-amplitude signals.

t-SNE
t-distributed Stochastic Neighbor Embedding; a non-linear dimensionality reduction technique for visualizing high-dimensional embeddings in 2D.

Latent embedding
The internal 64-dimensional representation produced by the FNO projection MLP before classification.

AdamW
A variant of the Adam optimizer with decoupled weight decay, providing better regularization.

Cosine annealing
A learning rate schedule that smoothly decays the learning rate following a cosine curve.

14. Further Reading

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